

CHEME 498

Final Project

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Introduction

- This research aims to study the performance of gas sensor classification at extremely low concentrations and compare the traditional vs. new approach for GSA classification.
 - Concentration below the minimum detection threshold
 - Difficulty of classifying gas sensor array data due to its high-dimensional features and small sample sizes

Dataset

- We utilized a GSA dataset containing a total of 9000 responses from 10 sensors to 6 different gases.
 - The dataset was split into train and test sets, with test being 20% of your data with a random state of 42. Standard scaler was implemented so that the data has unit variance.

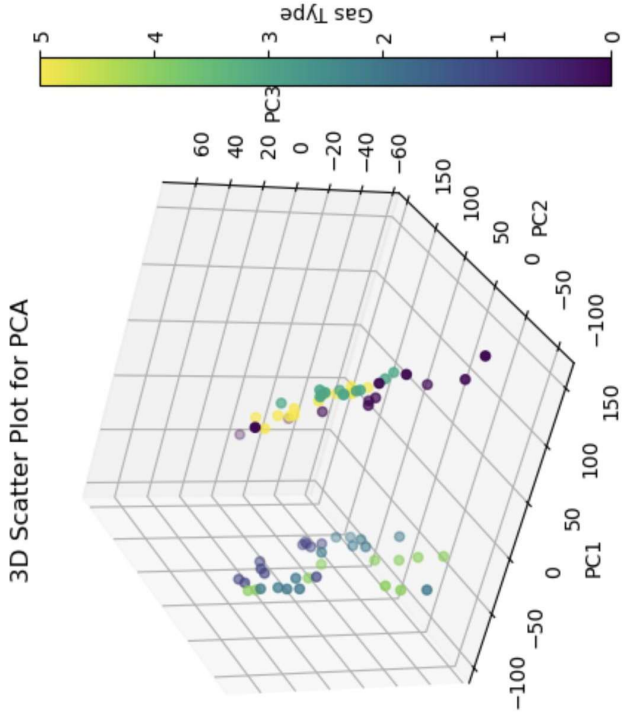
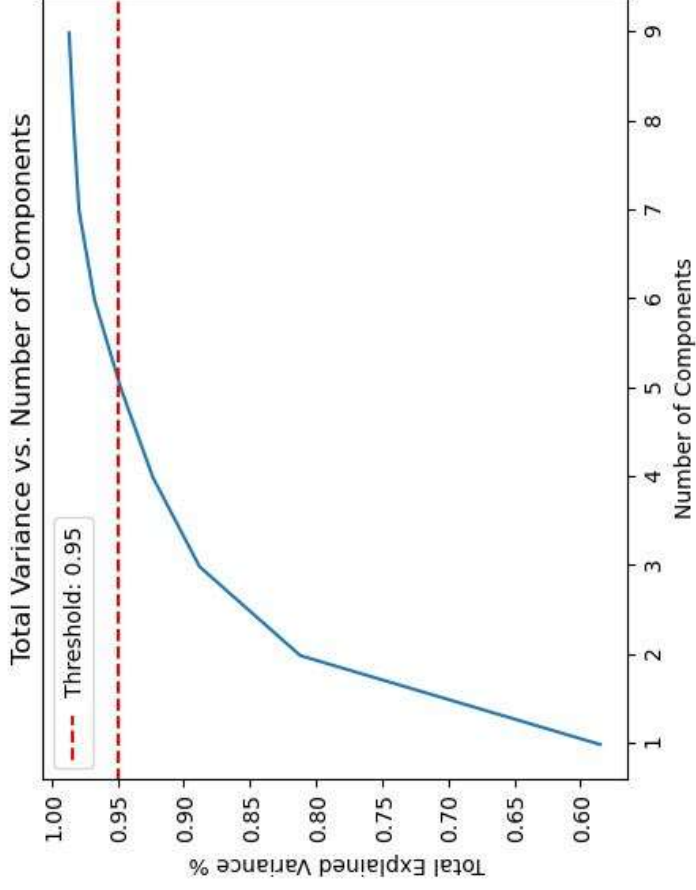
	ethanol	100ppb	0.3565	0.3345	0.3575	0.333	0.3565.1	0.3345.1	0.3565.2	0.3355	...
24	acetone	50ppb	0.3500	0.3320	0.3525	0.3355	0.3280	0.3330	0.3515	0.3320	...
45	ethyl acetate	100ppb	0.3695	0.3565	0.3675	0.3685	0.3675	0.3540	0.3710	0.3540	...
0	ethanol	100ppb	0.3525	0.3305	0.3525	0.3320	0.3550	0.3305	0.3525	0.3280	...
15	acetone	100ppb	0.3515	0.3305	0.3515	0.3330	0.3515	0.3320	0.3515	0.3305	...
68	isopropanol	200ppb	0.3490	0.3280	0.3500	0.3295	0.3500	0.3320	0.3490	0.3280	...

5 rows x 9002 columns

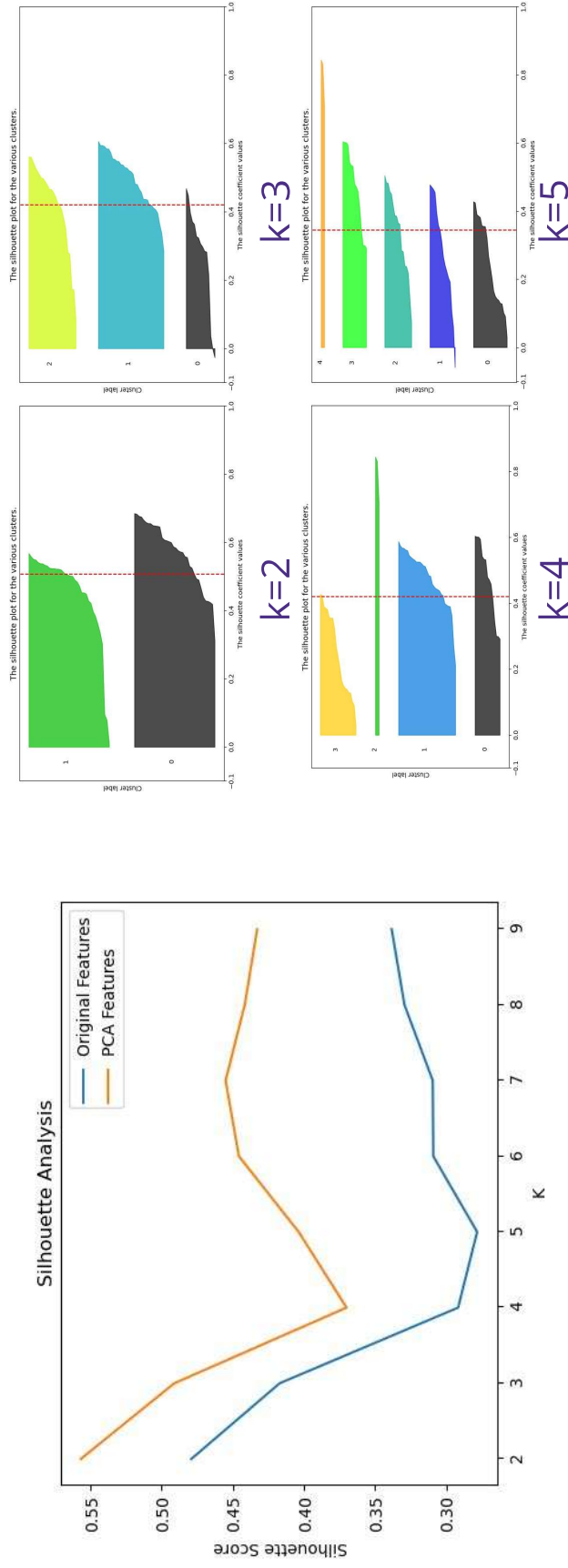
Feature Dimensionality Reduction

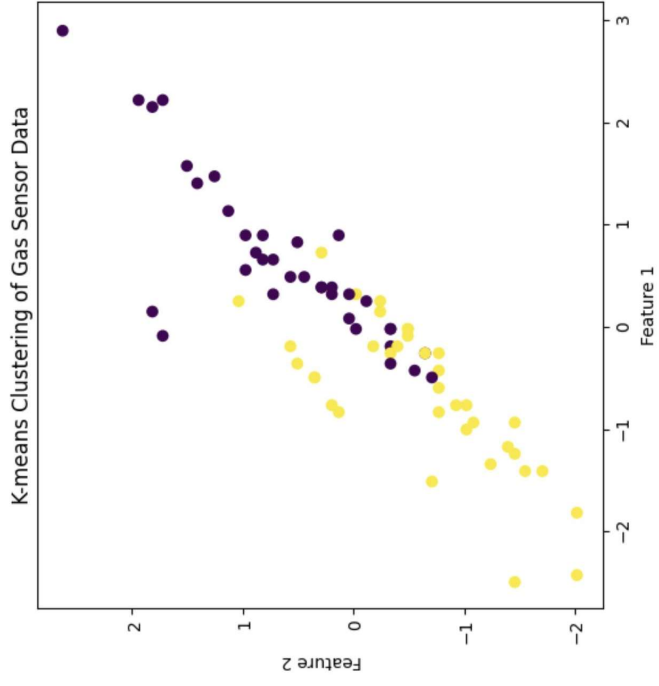
- Due to the high-dimensional features, we needed to reduce the dimensionality by implementing PCA before classification.
 - Reduce overfitting
 - Remove redundant features
 - Reduce computational time
- However, some features that are crucial for classification might also be removed, resulting in less accurate results.

- Total explained variance was calculated at different numbers of components. A 95% variance threshold was chosen to determine the most optimal component.

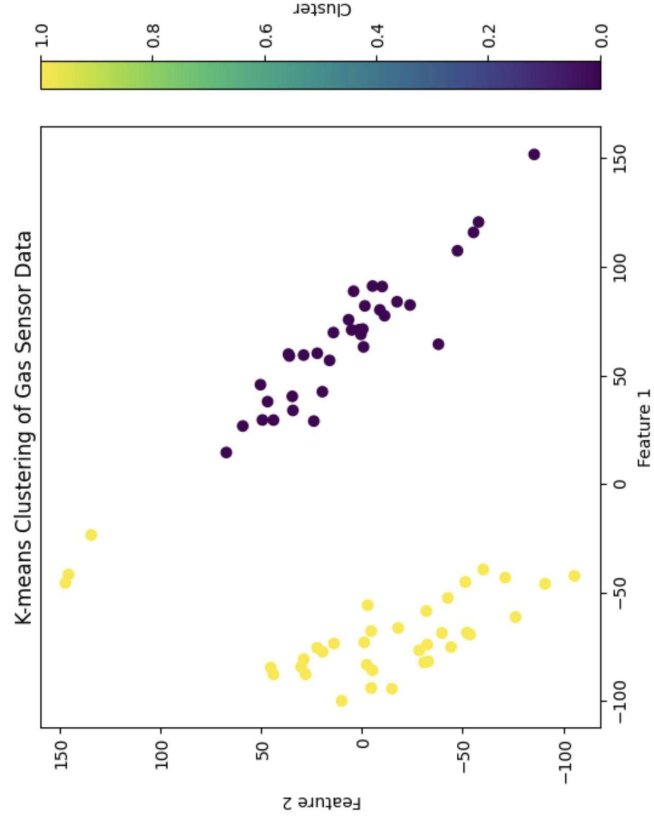


- To test the effect of dimensionality reduction , we implemented K-means with and without PCA features to see how it affects the clustering performance.
- Again...what's the most optimal number of clusters?





No PCA

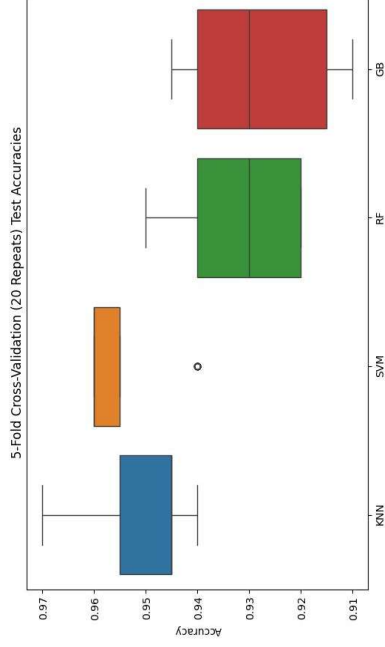


After PCA

PCA improves the clustering performance of K-means.

Classifier Implementation

- Some classic classifiers we utilized were KNN, SVM, RF, and Gradient boosting. Each classifier was carried out 5 fold cross validation 20 times to generate a comparative boxplot of test accuracies for the most optimal model.



Potential Issues

- Time consuming
- Overfitting due to high-dimensional features and small sample size of GSA data

Overall, approaching the GSA data this way results in poor classification performance. That's why the research team proposed a new method called **feature ensemble-based extreme learning machine framework (FE-ELM)** in the paper.

Traditional

Feature Dimensionality
Reduction (e.g. PCA)



Classic Classifier
(e.g. KNN, SVM)

FE-ELM

Downsampling



Formed fused feature
subsets



Implementing base ELM
on each subset

FE-ELM

- Inspired by the idea of **ensemble learning** methods
 - Train multiple weak classifiers and integrate them to a strong classifier, improving overall learning performance.
- Solves the overfitting problem!
- GSA data can be directly used without prior PCA.



THANK YOU!

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